Unified Swarm Operation SDD

8-Robot Heterogeneous Squadron (Go2 + Hub UGV + Deputy UGV + 1~5 Follower UGV)

System Design Document

문서번호 SAN-SDD-SWARM-001

v1.5

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Related Docs	SAN-IDS-CMD-001, SAN-TST-INT-001, SAN-OPS-SOP-001
Message Format	Google Protobuf v3 + ROS 2 .msg/.srv/.action
Middleware	ROS 2 Humble + DDS (Fast-DDS / Cyclone)
Communication	Wi-Fi 6 Mesh (OpenWrt) + Military LTE + LoRa

Revision History

Version	Changes
v1.0	Initial release. Swarm operation core architecture (Go2 Leader + Hub UGV + 7 Follower).
v1.1	Integrates the following changes: (1) 360° surveillance distribution — Leader 10s-period sector assignment, Follower front 180°, Hub UGV rear 180°; (2) Follower Pan-tilt sweep at 30°/sec during driving; (3) Hub UGV dual SBC (SLAM-dedicated + Comm/Video-dedicated); (4) SLAM collection period 1Hz → 30-60s aggregated; (5) Video streaming GStreamer UDP/SRT, Android APP pull mode; (6) Wi-Fi 6 Mesh OpenWrt 24.10 applied; (7) UGV LiDAR Robosense E1 adopted (Leader Go2 retains Unitree L1). This document absorbs SAN-SDD-SUR-001 v1.0 supplementary SDD.
v1.2	PM decisions D1~D7 + A1~A5 applied: (D1) Squadron Follower loss (8 → 7 followers re-mapped); (D2) Leader succession deputy_chain = [2, 3, 4, 5, 6, 7, 8], Hub UGV(S2) as priority 1 (legacy v1.2 stance, later reversed by v1.4); (D3) CycloneDDS NIC binding specified; (D7) AI inference Phase 1 RK3588 / PoC Hailo M.2.
v1.3	Cost Map + SLAM 5s + LTE Redundancy: (1) §9 SLAM aggregation period 30-60s → 5 sec; (2) §4 Robosense E1 VFOV 25° → 90° correction; (3) §4.5 UGV spec table newly added (1300×850mm, 12 km/h initially, slope 30°, obstacle 235mm, ditch 220mm); (4) LTE redundancy — Follower S3 LTE backup activates on Hub LTE failure.
v1.4	Deputy UGV pre-designation + 4-tier Leader succession + Limp Mode: (1) §5.5 Hub-Deputy redundancy — Deputy UGV (S3) takes over LTE + SLAM aggregation + GStreamer SRT relay on Hub failure; (2) §5.6 4-tier Leader succession (Deputy 1st, Hub 2nd, max-battery follower 3rd, Limp Mode 4th); (3) §5.7 Limp Mode — Hub+Deputy both fail, Android mesh-direct control retained including fire/strike commands; (4) Deputy UGV has identical hardware to Hub UGV (dual SBC + LTE).
v1.5	**DCN-2026-001 applied (2026-05-12): (1) All UGVs (S2~S8) ship with LTE modem standard — lte_backup_chain extended to [3, 4, 5, 6, 7, 8]; (2) UGV max speed 12 km/h → 10 km/h corrected to match AVTBOT TinS-13 chassis hardware spec; Cost Map look-ahead 2.1 s → 2.5 s (still within 5 s KPP); (3) Squadron size unified — 8 robots maximum / 4 robots minimum (test). SAN_SwarmOperation_Integrated v1.0 ARCHIVED; (4) Limp Mode fire authorization policy confirmed (Option A, SDD §5.7.2) — fire retained via Android mesh-direct HMAC-SHA256 with two-key arming + red banner + audit log. OPS §7 aligned; (5) Standard toolchain explicit (Ubuntu 22.04 + ROS 2 Humble C++ + GStreamer 1.20+) at §10.1.1; (6) §1.2 body "9-robot" → "8-robot" 일괄 정정; (7) §1.2.1 Squadron Composition table re-structured with Deputy UGV row + Follower 1-5 variable count.**

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1. Overview

1.1 Purpose and Scope

This document is the integrated System Design Document for the SAN swarm-controlled small tactical robot platform. It absorbs the v1.0 core operational framework and the seven additional change items confirmed in v1.1 (see Revision History) so that the entire system design can be understood from this single document.

1.2 System Overview

The system is an 8-robot heterogeneous swarm (maximum size; 4-robot minimum is supported for test configurations) that absorbs forward risk for an infantry squad while providing counter-drone defense, reconnaissance, and engagement support. A single Android tablet operated by one soldier commands the entire formation.

1.2.1 Squadron Composition

Role	Qty	Platform	Primary Mission
Leader	1	Unitree Go2 Pro/Edu Plus + RK3588 (with camera) + Wi-Fi 6 Mesh radio	Forward point, recon path lead, 1-sec prediction broadcast
Hub UGV (S2)	1	330 kg tracked UGV (dual SBC): Robosense E1 + LTE + RK3588 #1 (camera) + RK3588 #2 + Wi-Fi 6 Mesh	SLAM aggregation (5 s), comm hub, primary LTE gateway, Leader succession 2nd priority
Deputy UGV (S3) ★v1.4	1	330 kg tracked UGV (dual SBC): identical hardware to Hub UGV (Robosense E1 + LTE + RK3588 #1 + RK3588 #2 + Wi-Fi 6 Mesh) for hot-swap takeover	Standby; Leader succession 1st priority, Hub takeover 1st priority (LTE + SLAM + GStreamer SRT relay all transferred)
Follower UGV (S4~S8)	1~5 (variable)	330 kg tracked UGV (single SBC): Robosense E1 + LTE + RK3588 (with camera) + Wi-Fi 6 Mesh	V leg + rear guard, RCWS / EO·IR / RF jammer payload

1.2.2 Key Performance Parameters (KPP)

KPP	Target	Achievement Strategy
Formation hold avg error	≤ 2 m	1-sec prediction broadcast + 5-Tier auto avoidance
Proximity hazard reaction	≤ 300 ms	Tier 1.5 auto reroute + Nav2 local planner
Swarm control latency	≤ 150 ms	Wi-Fi 6 Mesh (OpenWrt) + DDS RELIABLE QoS
Leader takeover reform	≤ 10 sec	Modified Raft + Hub UGV priority succession
Assembly success rate	≥ 95%	Hungarian slot assignment + smooth transition
Drone detection range (RF)	≤ 2 km	RF scanner + 360° surveillance distribution
Real-time video transmission	≤ 200 ms	GStreamer SRT live latency 120ms tune
Target classification accuracy	≥ 90% (≤ 2s)	AI inference + 360° Pan-tilt sweep
Pan-tilt tracking error	≤ 0.05° (500m, 3m/s)	IMU-based active stabilization
Firing accuracy	≥ 90% (150m, 45cm)	Predicted lead point + gyro stabilization

2. System Architecture

2.1 V-Formation Standard Layout

```
                                   Leader (Go2)
L                                         |                        F1  ------------
+-------------  F2                        (front-left)    |             (front-right)
|                  F3  -------------------+-------------------  F4                 (left
flank)           |                     (right flank)
|              F5 -----------------------+----------------------- F6             (left
rear-flank)          |                            (right rear-flank)
|                                         H <- Hub UGV (center-rear, dual SBC)
|                                         F7 (rear guard)
v                                    Operator Tablet
(Android APP)
```

2.2 Triple-Redundant Communication

Channel	Protocol	Range	Latency	Bandwidth	Purpose
Primary (Wi-Fi 6 Mesh)	802.11ax (OpenWrt 24.10)	~50m (mesh)	5-20 ms	100-300 Mbps	Intra-swarm real-time
WAN (Military LTE)	LTE Mil band	~5 km	10-50 ms	50-100 Mbps	C2, video backhaul (via Hub)
Emergency (LoRa)	920 MHz ISM	~10 km	100-500 ms	0.3-50 Kbps	E-stop, beacon

2.3 ROS 2 + DDS Middleware Stack

```
     +-----------------------------------------+     |  Application Layer  (BT,
formation cmds) |     +-----------------------------------------+     |  Message Layer
(Protobuf / ROS msg) |     +-----------------------------------------+     |  DDS Layer
(Fast-DDS / Zenoh)    |     +-----------------------------------------+     |  Security
Layer     (DDS-Security AES-256)|     +-----------------------------------------+     |
Transport Layer    (UDP / TCP)           |
+-----------------------------------------+     |  Physical Layer     (Wi-Fi 6 / LTE /
LoRa)|     +-----------------------------------------+
```

3. Hub UGV — Dual SBC Architecture

3.1 Rationale for Dual SBC

v1.0 specified a single RK3588J SBC for the Hub UGV. In v1.1, the following concurrent requirements clearly exceed the single-SBC budget:

- 8-robot SLAM data fusion (multirobot_map_merge + g2o pose-graph)
- Wi-Fi 6 Mesh routing oversight (OpenWrt batman-adv monitoring)
- Military LTE gateway (mwan3 failover)
- Video stream forwarding (GStreamer SRT/UDP pipeline)
- DDS Security encrypt/decrypt (AES-256-GCM)
- 360° surveillance dispatch oversight

3.2 Dual SBC Role Split

Aspect	SBC #1 (SLAM)	SBC #2 (Comm/Video)
Model	Jetson Orin or RK3588J	RK3588J
Memory	16 GB LPDDR5	8 GB LPDDR5
Storage	256 GB NVMe	128 GB NVMe
Primary task	SLAM aggregation fusion (g2o)	Wi-Fi6 Mesh + LTE gateway
Secondary task	Global map broadcast	GStreamer SRT/UDP forwarding
Processes	slam_toolbox, multirobot_map_merge	OpenWrt monitor, GStreamer, IPsec/VPN
Network	Internal GbE (direct to SBC#2)	Wi-Fi 6 + LTE modem + Internal GbE
Power	Main battery 12V rail	Main battery 12V rail
Cooling	Independent heatsink + fan	Independent heatsink + fan

3.3 SBC Interconnect Bus

```
     +-----------------------------------------------------+     |            Hub UGV
(internal layout)                |     |
|     |  +--------------+ GbE    +--------------+            |     |  |   SBC #1     |
<------>|   SBC #2     |            |     |  |   (SLAM)     |        |  (Comm/Video)|
|     |  |              |        |              |            |     |  | ROS2 nodes:  |
| ROS2 nodes:  |            |     |  | - slam_fuse  |        | - mesh_mon   |
|     |  | - map_pub    |        | - lte_gw     |            |     |  | - g2o_opt    |
| - gst_relay  |            |     |  |              |        | - dds_sec    |
|     |  +--------------+        +--------------+            |     |
|                        |     |                              v
```

```
|     |  Wi-Fi 6 Mesh --- LTE --- All robots + Operator       |
+-------------------------------------------------------+
```

3.4 SBC Failure Response (Partial Operation)

Failure Scenario	Behavior	Mission Impact
SBC #1 (SLAM) failed	Aggregated SLAM suspended; each follower runs local SLAM	Mission continues (global map unavailable)
SBC #2 (Comm) failed	Other Wi-Fi 6 mesh robots assist routing; LTE/video stopped	Mission pause or SBC #2 reboot attempted
Both SBCs nominal	All functions normal	Full mission capability

3.5 Leader Succession Priority

On Leader (Go2) loss, the Hub UGV takes over with top priority. Rationale: (1) Dual SBC computing headroom, (2) LTE backhaul present, (3) Rear central position is stable, (4) Hardcoded priority 1 in Modified Raft.

```
     @dataclass     class CandidatePriority:         is_hub_ugv: bool         #
priority 1 (absolute)         battery_pct: int          # priority 2
sensor_health: int        # priority 3         comm_centrality: float    # priority 4
robot_id: int             # priority 5 (tie-breaker)
```

4. Sensors — LiDAR + EO/IR + RF

4.1 LiDAR Selection

Aspect	Leader (Go2)	Follower / Hub UGV
Model	Unitree L1 (built-in)	Robosense E1 (solid-state)
Mechanism	Multi-line mechanical	Solid-state, 905nm
Range	30 m (typical)	200 m (10% reflectivity)
FoV (horizontal)	360° (rotating)	120° (solid-state)
FoV (vertical) ★	±30°	**90°** (v1.3 correction, prior 25° typo)
Mount height ★	Chassis top	**500 mm above ground**
Scan start ★	Front 0 m	**250 mm forward of chassis** + upward scan
Point density	~200,000 pts/s	~921,600 pts/s
Rate	10 Hz	10 Hz (or 20 Hz)
Mass	~250 g (built-in)	~600 g
Unit cost (volume)	Included in Go2 package	~$200-500
Purpose	Leader 1s prediction + close avoidance	Long-range recon + SLAM + Cost Map + target ranging

★ = v1.3 new/corrected. The 90° vertical FOV is essential for simultaneous ground plane segmentation and upward obstacle detection (branches, vehicle undercarriage, etc.).

4.2 Robosense E1 Rationale

- **Solid-state**: no moving parts, robust against military vibration/shock
- **200m range**: long-range recon + cross-validation with RF scanner
- **120° FoV**: well-suited for UGV forward recon combined with Pan-tilt gimbal
- **600g mass**: minimal load on tracked UGV
- **Price**: $200-500 volume cost, competitive vs Mid-360 ($800-1200)
- **Korean supply**: Robosense Korea distributor (Kernel Robotics) availability

4.3 Optical Cameras (EO/IR)

Aspect	Daytime EO Camera	Thermal IR Camera
Resolution	FHD 1920x1080, 30x optical zoom	640x512
Rate	30 fps	30 fps

Aspect	Daytime EO Camera	Thermal IR Camera
HFOV (Wide)	~60° (1x)	~24°
HFOV (Tele 30x)	~2-3°	~0.8-1.2°
Mount	Follower/Hub Pan-tilt gimbal	Follower/Hub Pan-tilt gimbal
Purpose	Daytime recon, target ID	Night/smoke environment
Night operation	Secondary	Primary asset

4.4 RF Scanner + Jammer

- **RF scanner**: 2.4 / 5.8 GHz / GNSS L1·L2 — drone detection ≤ 2 km
- **RF jammer**: 2.4 / 5.8 GHz, 20W class — effective radius ≥ 1 km, soft-kill
- **Hub UGV gets primary mount** (high-gain antenna) + selected followers carry secondary units

4.5 UGV Platform Specifications (v1.3 new)

Follower UGV and Hub UGV share the same tracked platform (Hub UGV adds only dual SBC + high-gain RF antenna). This section provides the input parameters for Cost Map computation and obstacle avoidance algorithms. Unitree Go2 (Leader) exceeds these specs (better speed/slope capability).

Parameter	Specification	Note / Usage
Chassis dimensions	Length 1300 mm × Width 850 mm	Cost Map inflation radius input
Max speed	10 km/h (2.78 m/s)	Cost Map look-ahead distance input
Slope capability	30°	Traversability layer lethal threshold
Obstacle clearance height	235 mm	Lethal obstacle classification threshold
Ditch / gap clearance width	220 mm	Lethal ditch classification threshold
LiDAR mount height	500 mm above ground	Robosense E1 (§4.1)
LiDAR scan start point	250 mm forward of chassis + upward	Close-range avoidance + overhead obstacles

4.5.1 Cost Map Derived Values

Derived value	Formula	Result
7m reach time	7 m / 2.78 m/s	2.5 sec (predictive window, v1.5 recalculated: 7 m /

Derived value	Formula	Result
		2.78 m/s ≈ 2.52 sec)
Cost Map coverage distance	max(7 m, v × 2.5 sec margin)	**>= 7 m** (new KPP)
Cost Map update latency	Sensor input -> cost map publish	**<= 5 sec p99** (new KPP)
Inflation radius	$\mathrm{sqrt}(1.3^2 + 0.85^2) / 2 + 0.25$ safety	~1.0 m
Cost Map resolution	0.05 m (5 cm)	235mm obstacle -> 4-5 cells
Cost Map area	Forward 7 m + side/rear margin	14 × 14 m (280 × 280 cells)

5. Wi-Fi 6 Mesh — OpenWrt 24.10

5.1 OpenWrt Selection Rationale

Aspect	Stock firmware	OpenWrt 24.10 (adopted)
Mesh capability	Vendor EasyMesh	802.11s + batman-adv + prplMesh
QoS	Basic WMM	SQM cake/fq_codel + nftables DSCP
Multicast (DDS) ctrl	Limited	Full + IGMP snooping (multicast_mode)
LTE Failover	Manual, limited	mwan3 automatic (< 8s detect)
Security patches	Quarterly/annual vendor lock	Monthly immediate
Military verification	Vendor black-box	Open-source SBOM verifiable
WireGuard VPN	Unavailable	Supported (Phase F+)
License cost	$0	$0 (GPL)

5.2 Hardware Candidates

Phase	Recommended Model	Notes
B-C (Lab/PoC)	GL.iNet Flint AX1800	OpenWrt 24.10 preinstalled, ~$110
D-E (Field PoC)	GL.iNet Flint 2 (MT6000)	Next-gen, 4x4 5GHz, ~$150
F+ (production)	Compex WPJ563 (industrial)	IPQ8074, industrial certified, ~$250
Spare	Xiaomi AX3000T	Korean direct purchase, ~$50

5.3 Mesh Topology

```
     Operator Tablet (Android)         | AP mode 5GHz (SAN-MESH-OPS)         v
[Mesh Router #1 (Controller, LTE backhaul)]          | 802.11s + batman-adv (BATMAN V)
|-[Mesh Router #2 (Agent)]          |       |          |       \- Robot 1-9 (Wi-Fi STA)
\-[Mesh Router #3 (Agent)]                  |                  \- Robot 1-9 (Wi-Fi STA)
```

5.4 Key OpenWrt Configuration

Setting	Value
OS	OpenWrt 24.10 stable
Mesh PHY	802.11s, 5GHz CH36, HE40 (40 MHz)
Mesh Auth	WPA3-SAE
Mesh ID	san-mesh-001
L2 Routing	batman-adv BATMAN_V
AP Steering	DAWN (decentralized WLAN controller)
Multicast tuning	multicast_mode=1, igmp_snooping=1
WAN Failover	mwan3 (WAN <-> LTE, 8s detect)
QoS	SQM cake + nftables DSCP marking
DSCP marking	FollowerTarget->EF, RobotStatus->AF31, GStreamer->AF21
Security	WPA3-SAE + WireGuard (Phase F+)
Country code	KR (EIRP <= 23 dBm)

5.5 Hub-Deputy Redundancy — Hub UGV Failure Backup (v1.4 expanded)

v1.4 change: v1.3 only defined LTE backup (lte_backup_chain [3, 5]). v1.4 introduces **pre-designated Deputy UGV (S3)** that takes over **all Hub UGV roles** (LTE + SLAM aggregation + video relay). Deputy UGV has **identical hardware** to Hub UGV (dual SBC + LTE modem) for immediate replacement.

Hub UGV (S2) SBC #2 is the primary LTE gateway. On SBC #2 failure, LTE modem failure, or Hub UGV total loss, the LTE gateway role auto-promotes per the ordered chain **[3, 4, 5, 6, 7, 8]** (v1.5: extended to all UGVs since every UGV ships with an LTE modem). Order: **Deputy UGV (S3) 1st** (full Hub takeover — LTE + SLAM aggregation + video relay all transferred), then Followers S4→S5→S6→S7→S8 (LTE-only promotion — SLAM aggregation not transferred due to single SBC budget). Pre-designated backup with split-brain prevention via monotonically increasing lte_term.

5.5.1 Deputy UGV Spec (Complete Hub Backup)

Item	Hub UGV (S2)	Deputy UGV (S3) ★v1.4 new
robot_id	2	3
Dual SBC	RK3588 #1 (SLAM) + #2 (Comm)	RK3588 #1 (SLAM) + #2 (Comm)

Item	Hub UGV (S2)	Deputy UGV (S3) ★v1.4 new
LTE modem	Mounted (primary)	Mounted (backup)
High-gain RF ant.	Mounted	Mounted
Primary role	LTE + SLAM aggregation + video relay	Leader succession 1st + Hub backup
Failover scope	—	All: LTE + SLAM + video relay
YAML	is_hub_ugv: true	is_deputy_ugv: true
Promotion threshold	—	Hub heartbeat missing 5 sec
Term increment	—	hub_term += 1 (split-brain prevention)

★ v1.4 new: Upgraded from v1.3's LTE-only single backup to **complete Hub function backup**. Deputy UGV has identical hardware to Hub UGV with swappable OS image.

5.5.2 Hub -> Deputy Transition Scenario (v1.4 expanded)

```
     t = 0       Hub UGV (S2) LTE backhaul nominal                 operator <-> Hub(S2)
<-> swarm          t = T0      Hub UGV (S2) failure detected (heartbeat 5s missing or
both SBC fail)                Deputy UGV (S3) detects hub_heartbeat_timeout          t
= T0+5s   Deputy S3 decides Hub takeover + broadcasts:
HubRoleAnnouncement(role=HUB_PROMOTED, hub_term=N+1)
LTERoleAnnouncement(role=LTE_PROMOTED, lte_term=M+1)          t = T0+7s   Deputy S3
takeover actions:                 - LTE modem activation (mwan3 weight 100, via libuci)
- SLAM aggregation takeover (multirobot_map_merge start)                - GStreamer
relay takeover (SRT listener activate port 8888)                -
RobotStatus.lte_active = true, is_hub_role = true publish          t = T0+10s  Operator
display: "Hub UGV failure — Deputy UGV (S3) took over"                operator <-> S3
<-> swarm path switch (LTE + Wi-Fi 6 mesh)                        *On Hub S2 recovery: S3
-> HUB_DEMOTED, Hub S2 -> HUB_PROMOTED, hub_term += 1     *On Deputy also failing: §
5.7 Limp Mode entered (Wi-Fi 6 only)
```

5.5.3 Deputy UGV Verification Policy

- Deputy SBC #1, #2 hardware presence check every 30 sec (same as Hub)
- LTE modem standby health check every 5 min (no activation, small ping)
- SLAM aggregation shadow mode — Deputy receives Hub's aggregated result for validation (consistency >= 95%)
- Operator screen displays both Hub/Deputy link health + battery levels
- **On both Hub + Deputy failure: enter § 5.7 Limp Mode** (Wi-Fi 6 only)

5.6 4-Tier Leader Succession (v1.4 new)

When Leader robot (Unitree Go2, S1) fails, succession proceeds per a 4-tier priority. v1.4 replaces v1.3's simple deputy_chain[2,3,...,8] with **role-based prioritization**.

5.6.1 Leader Succession Priority

Priority	Candidate	Condition	Term handling
1st ★	**Deputy UGV (S3)**	Both Deputy SBCs healthy + battery >= 20%	leader_term += 1
2nd ★	**Hub UGV (S2)**	Deputy failed + both Hub SBCs healthy + battery >= 20%	leader_term += 1
3rd	**Max-battery follower**	Both Deputy + Hub failed + at least 1 operational follower	leader_term += 1
4th (Limp)	**Mission ended / wait**	0 operational followers or all robots battery < 10%	Safe mission termination

★ v1.4 new: Deputy UGV has higher Leader succession priority than Hub UGV because (1) Hub UGV handles SLAM aggregation + video relay (heavy load), adding Leader role is risky; (2) Deputy UGV is in standby with ample processing margin; (3) Hub continues to operate SLAM/LTE.

5.6.2 Succession Scenario

```
     t = 0       Leader (Go2 S1) operating normally                 Deputy S3, Hub S2
both in standby          t = T0      Leader heartbeat 200ms missed 7 times in a row
(1.4 sec)                 Deputy S3 detects first (tracking leader_term in shadow mode)
t = T0+2s   Deputy S3 decides Leader takeover:
LeaderRoleAnnouncement(role=LEADER_PROMOTED, leader_term=N+1,
reason="leader_heartbeat_timeout") broadcast          t = T0+3s   Deputy S3 executes
Leader duties:                 - 1-sec predicted position broadcast at 10Hz
- Hungarian algorithm (formation ID assignment)                 - Hub UGV continues
SLAM aggregation (role separation)          t = T0+5s   Operator display: "Leader
failure — Deputy UGV (S3) took over Leader"          *On Leader S1 recovery: Deputy ->
LEADER_DEMOTED, Leader -> LEADER_PROMOTED, leader_term += 1     *On Deputy also
failing: 2nd-priority succession to Hub UGV (S2)     *On all 3 failing: max-battery
follower -> § 5.7 Limp Mode simultaneously
```

5.6.3 Battery-based Follower Succession (3rd-tier)

When both Deputy + Hub failed, the operational follower with the **max battery percent** takes over Leader. Each robot publishes RobotStatus with battery_percent every 200ms. Candidate robots broadcast LeaderRoleAnnouncement(candidate); highest battery_percent (tiebreaker: lowest robot_id) is promoted.

Condition	Action
follower battery >= 50%	Leader candidate group 1 (normal operation)

Condition	Action
follower battery 30-50%	Leader candidate group 2 (time-limited missions)
follower battery 10-30%	Leader candidate group 3 (emergency RTB only)
All robots battery < 10%	Abandon Leader succession -> § 5.7 Limp Mode

5.7 Limp Mode — Hub+Deputy Both Failed (v1.4 new)

When **both Hub UGV (S2) and Deputy UGV (S3) are inoperable**, SLAM aggregated distribution and LTE external control are unavailable. Operation falls back to **Wi-Fi 6 mesh-based robot control** (Limp Mode) — a reduced operational mode.

5.7.1 Limp Mode Entry Conditions

Condition	Result
Hub UGV (S2) heartbeat missing 5 sec	Deputy auto-takeover attempt
Both Hub + Deputy heartbeat missing 5 sec	**Enter Limp Mode**
Hub heartbeat normal but LTE modem + Deputy also failed	**Limp Mode (LTE-only loss)**
Wi-Fi 6 mesh also disconnected	**Total Outage** (mission abort)

5.7.2 Limp Mode Operational Features (★ v1.4 MODIFIED)

v1.4 key revision: Even in Limp Mode, the Android app connects directly via Wi-Fi 6 mesh and **maintains full control of the surviving swarm** — fire/strike commands + video viewing both available. LTE deactivation affects only LTE-backbone external remote access.

Function	Normal Operation	Limp Mode (Hub+Deputy failed)
Wi-Fi 6 mesh 802.11s	✅ Active	✅ **Only comm path**
Android app operator control	Via LTE backbone	✅ **Wi-Fi 6 mesh direct connection**
LTE external control (remote)	✅ Operator remote access	❌ Unavailable (Hub/Deputy LTE absent)
Global SLAM aggregation	✅ 5-sec period	❌ Unavailable (per-robot local SLAM only)
Cost Map avoidance (local)	✅ 1Hz	✅ **Continues** (per-robot computation)

Function	Normal Operation	Limp Mode (Hub+Deputy failed)
1-sec predicted position broadcast	✅ 10Hz	✅ Active (Leader-succession robot publishes)
GStreamer SRT video relay (via Hub)	✅ Hub SBC #2	❌ Hub absent
GStreamer UDP/SRT direct video	Via Hub	✅ **Each robot sends directly to Android**
RTSP video	Debug only	✅ Secondary path (debugging)
Priority 1 commands (EmergencyStop)	LTE + mesh multipath	✅ mesh delivery (200ms reliable)
Fire/strike commands	✅ Production mode only	✅ **Available via Android mesh-direct HMAC-SHA256 auth** (v1.5 D-004 Option A confirmed; Two-key arming + red banner + audit log required — see §5.7.2.1)
Mission continuation	Full missions	🟡 Simple missions (recon, movement, fire)
Complex missions (formation change, AI)	✅ Normal	🟡 Paused (single-SBC processing burden)

5.7.2.1 Limp Mode Fire Authorization Policy (★ v1.5 confirmed — DCN-2026-001 D-004)

Per DCN-2026-001 D-004, fire authority is RETAINED in Limp Mode under the safeguards below. This overrides the conservative reading some early v1.4 OPS drafts considered. OPS §7 has been aligned to this policy.

Safeguard	Implementation
Authentication	Android tablet ↔ surviving robot via Wi-Fi 6 Mesh, HMAC-SHA256 with per-mission shared secret. LTE bypass.
Two-key arming	Operator screen requires (1) target tap = single-key confirm + (2) “Confirm Fire” red button = second-key arm. Limp Mode displays “LIMP MODE — fire permitted, situation awareness reduced” red banner above firing UI.
Audit logging	Every Limp Mode fire event tagged limp_mode_fire = true in audit log + broadcast to all surviving robots via OperationState.last_limp_fire_ms field.
Suspended functions	Complex missions (formation change, AI inference scheduling, autonomous engagement) remain paused per §5.7.2. Only operator-initiated fire is permitted.

Safeguard	Implementation
Recovery transition	On Hub or Deputy recovery → LIMP_MODE_EXITED alert + operator must explicitly re-enable autonomous engagement.

5.7.3 Limp Mode Automatic Actions (★ v1.4 MODIFIED)

1. **Wi-Fi 6 mesh routing optimization** — exclude Hub/Deputy nodes, redistribute routing to other robots
2. **Android app connection auto-switch** — Android app connects via Wi-Fi 6 mesh directly (via nearest surviving follower)
3. **Video send mode auto-switch** — each robot's GStreamer UDP/SRT pipeline sends directly to Android app IP (bypassing absent Hub SRT relay)
4. **Mesh direct auth activation** — fire/strike commands authenticated via HMAC-SHA256 mesh shared secret (LTE-bypassed)
5. **Complex missions paused** — only formation change / AI inference paused; simple missions + fire remain operational
6. **Audit log enhanced** — all Limp Mode entry/exit events transmitted with priority over mesh

5.7.4 Limp Mode Exit (Recovery) Conditions

- Hub UGV or Deputy UGV — at least one heartbeat recovers
- Recovered robot takes over SLAM aggregation (within 5 sec)
- LTE modem reactivation broadcasts LTERoleAnnouncement
- Operator screen: "Normal operation resumed"

5.8 IP Address Allocation

Subnet	Purpose
192.168.42.0/24	Mesh + clients (DHCP pool)
192.168.42.1	Router #1 (controller + LTE gateway)
192.168.42.2-9	Router #2-3 (agents)
192.168.42.10	Hub UGV SBC #2 (mesh ingress/egress)
192.168.42.11	Leader (Go2)
192.168.42.12-19	Follower UGV (F1-F5)
192.168.42.100-199	Android tablet + transient clients

6. Behavior Tree + 5-Tier Autonomous Avoidance

6.1 Mission BT Top-Level Structure

```
     Root: Fallback     +-- EmergencyHandler (P0)     |   +-- EmergencyStopRequested?
|   +-- StandAndPublishStop     |   \-- WaitForRelease     +-- ManualOverride (P1)
|   +-- InManualMode?     |   \-- ForwardManualCmdVel     +-- DegradedHealth (P2)     |
+-- HealthCritical?     |   \-- ReturnToHomeOrStand     +-- BatteryCritical (P3, <=30%
triggers Hub UGV takeover)     |   +-- BatteryLowEmergency?     |   \-- ImmediateStand
or HubTakeover     \-- NormalMissionFlow (default)         +-- PreCheck (PTP, RTK,
SLAM, Sensors)         +-- ResolveRole (Leader / Hub / Follower)         +--
ExecuteRoleAction         |   +-- If Leader:         |   |   +-- PlanGlobalPath (Hybrid
A*)         |   |   +-- PublishBreadcrumb (1m/0.5s)         |   |   +--
Publish1sPrediction (FollowerTargetMessage, 10Hz)         |   |   +-- ManageFormation
(Hungarian)         |   |   \-- DispatchSurveillanceSectors (10s period) * v1.1 new
|   +-- If Hub UGV:         |   |   +-- (Leader actions when in succession)         |
|   +-- Aggregate SLAM (SBC #1, 30-60s period)         |   |   +-- Broadcast Global Map
(SLAMAggregatedMap)         |   |   +-- Forward Video Stream (SBC #2)         |
|   \-- Survey Rear 180° (Pan-tilt sweep)         |   \-- If Follower:         |
+-- TrackLeaderPredictedPose         |       +-- ComputeOffsetByFormation         |
+-- ApplyTierAwareMotionLimit         |       \-- ExecuteSurveillanceSector (Sweep
30°/s) * v1.1 new         +-- CheckTier (T0->T4 transitions)         +--
ApplyTierAction         \-- PublishProgress
```

6.2 5-Tier Autonomous Avoidance

Tier	State	Trigger	Action
T0	PREDICTIVE_TRACK	1s prediction received OK	Track predicted position
T1.5	AUTO_REROUTE (new)	Obstacle in front of predicted position	±2m lateral detour
T1	NORMAL	No prediction, comm OK	Body-frame offset PID tracking
T2	CATCH_UP	$\delta > 1.5\ d_0$	1.2x speed
T3	HARD_CATCH_UP	$\delta > 2\ d_0$	Max speed
T4	BREADCRUMB_RECOVERY	$\delta > 4\ d_0$ or 60s comm loss	Trace Leader breadcrumb backward

6.3 1-Second Prediction Broadcast

```
     t = 0       Leader computes 1s-future pose (Hybrid A* lookahead)
+--> FollowerTargetMessage (10Hz, /swarm/predict/follower_target)      t = 50ms    Hub
UGV receives (Wi-Fi 6 direct 1-hop)      t = 150ms   Follower receives (worst case,
multi-hop mesh)                  ^----- 850ms margin -----v      t = 1000ms  Follower
arrives at predicted position                 T1.5 trigger: ±2m lateral detour, rejoin
on next cycle
```

6.4 Cost Map-based Obstacle Avoidance (v1.3 new)

Detailed algorithm for Tier 1.5 autonomous avoidance. Each UGV computes a Local Cost Map at 1 Hz on its own SBC (see §9.6), and uses Nav2 layered_costmap_2d + DWB local planner to perform ±2m lateral detours.

6.4.1 Avoidance Decision Flow

```
     Every 100 ms BT tick:       1. Receive follower_target (1-sec predicted position)
2. Query Local Cost Map (own SBC, 1 Hz refreshed)       3. Inspect predicted-path
cells:          - All cells cost < 50 (free)            -> stay T0          - Cell cost
>= 254 (lethal) detected    -> enter T1.5          - Cell cost 50~253 (inflated)
-> T1.5 (slow down)       4. On T1.5 entry:          - DWB local planner: goal = 1-sec
predicted pose          - costmap = static + obstacle + traversability + inflation
- Evaluate planner trajectory candidates -> pick lowest-cost          - Output
trajectory is typically a ±2m lateral detour       5. Publish command: /cmd_vel (from
Nav2 controller_server)       6. Confirm free space next cycle -> return to T0
```

6.4.2 Cost Thresholds (per UGV spec §4.5)

Condition	Cost	Avoidance Action
Obstacle height >= 235 mm (impassable)	254 (lethal)	**Mandatory detour**
Ditch width >= 220 mm (impassable)	254 (lethal)	**Mandatory detour**
Slope > 30° (impassable)	254 (lethal)	**Mandatory detour**
Obstacle 100~235 mm (risky)	200 (inflated)	Detour if possible, slow down
Slope 15~30°	100 (cautious)	Slow traverse
Inflation zone (footprint + 0.25 m margin)	150 (inflated)	Avoidance recommended
Slope < 15° + obstacle < 100 mm	0 (free)	Normal pass

6.4.3 Avoidance Failure Recovery

- **Tier 1.5 timeout (no free space within 5s)** -> Tier 2/3 forced traverse attempt
- **All trajectory costs >= 254** -> Stop + Backup behavior (0.5m reverse)
- **3 consecutive avoidance failures** -> Operator alert + "Manual intervention required" display

7. Formation (9 Types + Hungarian)

7.1 9 Formation Types

#	Formation	Use Case	Slot Computation
1	Column	Single column (narrow passage)	$y_{local} = -k \times d$
2	Line	Single rank (sweep)	$x_{local} = \pm k/2 \times d$
3	V-Shape	Recon default (60° standard)	$\theta=60°$ symmetric branches
4	Diamond	Omnidirectional perimeter	4 corners: $\pm d/\sqrt{2}$
5	Echelon-Left	Flank guard (left)	45° offset, $y=-k \times d/\sqrt{2}$
6	Echelon-Right	Flank guard (right)	Mirror of left
7	Box	Square (stationary perimeter)	4 corners + center
8	Vee-Inverted	Forward ambush	V with leader at rear
9	Free-Spread	Free dispersal (anti-drone task share)	Random within radius=d

7.2 4 Operational Presets (linked with SOP)

Preset	d	θ	Use	Selection Criterion
Narrow Penetration	3 m	40°	Narrow corridor, woods, alley	Passage width < 6m
Recon/Defense	5 m	90°	360° viewport, flank perimeter	Recon, patrol, base defense
Wide Recon	7 m	120°	Distributed surveillance	Open terrain recon
Assault	15 m	60°	Forward firepower + shell evasion	Enemy contact / breakthrough

7.3 Mission-Stage Auto Transitions (A → F)

```
    A. Assembly        ->  Free dispersal     B. Move-in          ->  Single column
C. Reconnaissance   ->  V-Shape (60°)     D. Engagement       ->  Single rank / V-Shape
E. Hold (objective) ->  Diamond     F. Withdraw         ->  Single column (reversed)
```

7.4 Hungarian Slot Assignment

```
    from scipy.optimize import linear_sum_assignment      def
assign_slots(current_poses, target_offsets):         """Min-cost: which robot -> which
slot?"""         n = len(current_poses)         cost = np.zeros((n, n))         for i,
pose in enumerate(current_poses):             for j, offset in
enumerate(target_offsets):                 target = (pose.x + offset[0], pose.y +
offset[1])                 cost[i][j] = math.hypot(target[0] - pose.x, target[1] -
pose.y)         row_ind, col_ind = linear_sum_assignment(cost)         return
dict(zip(row_ind, col_ind))
```

8. 360° Surveillance Distribution (v1.1 new)

8.1 Distribution Principles

The 8-robot heterogeneous squadron's 360° surveillance is partitioned by these principles. The Leader dynamically assigns sectors to each robot and refreshes every 10 seconds (or immediately on events).

Robot	Direction	Primary Mode	Role
Leader (Go2)	Front 0° ±30°	Fixed (front)	Recon + path clearance (no Pan-tilt)
F1 - F6 (Follower)	Front 180° (shared)	Sweep	Sweep within assigned sector
F7 (rear Follower)	Rear assist 30°	Sweep	Hub assist (optional)
Hub UGV	Rear 180°	Sweep	Primary rear surveillance + comm/SLAM hub

8.2 Front 180° Distribution (standard 8-robot)

Robot	Sector	Center	Primary Direction
Leader	-30° to +30°	0° (front)	Front 60°
F1	-90° to -30°	-60° (front-left)	Front-left
F2	+30° to +90°	+60° (front-right)	Front-right
F3	-120° to -60°	-90° (left flank)	Left flank
F4	+60° to +120°	+90° (right flank)	Right flank
F5	-150° to -90°	-120° (left rear)	Left rear flank
F6	+90° to +150°	+120° (right rear)	Right rear flank
Hub	+150° to -150°	+180° (rear)	Rear 180°
F7	Dynamic reinforce	Dynamic decision	Hub assist or threat sector

8.3 Pan-tilt Operation Modes

Mode	Pan Motion	Tilt Motion	Transition Trigger
Sweep	Reciprocating within sector	Horizon-based -5° to +10°	Default while driving
Fixed	Hold at commanded	Hold at commanded	Leader command or

Mode	Pan Motion	Tilt Motion	Transition Trigger
	direction	angle	operator
Track	Auto-follow target position	Auto-follow target position	AI detection or threat
Engage	Target lock-on + lead correction	Target lock-on + lead correction	Just before Confirm Fire

8.4 Sweep Speed

```
    ω_sweep = (sector_width - HFOV) × 2 / period      Example: sector_width = 90°,
HFOV = 60°, period = 6s              ω_sweep = (90 - 60) × 2 / 6 = 10°/sec
Example: sector_width = 180°, HFOV = 60°, period = 8s              ω_sweep = (180 - 60)
× 2 / 8 = 30°/sec      Standard sweep speed 30°/sec. Night thermal mode uses 15°/sec
(narrower HFOV).
```

8.5 Periodic Sector Refresh

Trigger	Refresh Period	Refresh Scope
Periodic refresh	10s (default)	Recompute all follower sectors
Formation change	Immediate	Reassign all sectors for new formation
Follower lost	Immediate	Redistribute 360° among remaining
Threat detected	Immediate (while present)	Priority assign to threat-side followers
Manual command	Immediate	Concentrate/disperse over specific area

8.6 Dynamic Adaptation Scenarios

8.6.1 Threat Sector Focus

When the RF scanner or a follower detects a threat, the Leader reassigns the 2-3 closest followers to the same threat sector. This yields (1) higher detection probability, (2) improved cross-cueing accuracy, (3) faster engagement.

8.6.2 Gap Filling

When one follower is lost (comm loss, power loss, etc.), the remaining followers automatically redistribute sectors. Transitioning 8 -> 7 robots increases average sector width proportionally while guaranteeing full coverage. The same algorithm scales for any squadron size from 8 down to 4 (minimum test configuration).

8.6.3 Recon vs Defensive Mode

Assault mode (offensive): narrow front sectors (15-20°/follower), wide rear sectors.
Defensive mode (point defense): balanced (45°/follower) front-rear.

9. SLAM Aggregated Collection

9.1 Rationale for Period Change (v1.0 → v1.1)

Aspect	v1.0 (1 Hz)	v1.1 (30-60s)	v1.3 (**5 sec**)
Comm load (8 robots)	80-320 KB/s	1-3 KB/s (1/100)	**6-20 KB/s (1/16)**
Hub SBC #1 load	Continuous 100% busy	Periodic 10-20% busy	**Periodic 40-60% busy**
Global map refresh	1s (excessive)	30-60s (recon adequate)	**5s (improved real-time)**
Real-time avoidance	Global dep. (poor)	Local SLAM (good)	Local SLAM + Cost Map
Bandwidth headroom	Low	Large headroom	Adequate for 1s pred + Cost Map

v1.3 rationale: requirement for improved real-time global map + Hub UGV dual SBC has sufficient processing power + alignment with Cost Map 1Hz refresh -> shortened from 30-60s to **5 sec**.

9.2 SLAM Data Flow

```
    [Each Robot] Local SLAM (slam_toolbox + Robosense E1, 1Hz)          |           |
Every 30-60s send aggregated occupancy grid delta          v     [Hub UGV SBC #1]
multirobot_map_merge          | - feature-based merge (ORB/RANSAC)          | - pose-
graph optimization (g2o)          v     [Hub UGV SBC #1] maintains global merged map
|           | Every 30-60s broadcast (or delta only)          v     [All Robots +
Operator Tablet] Nav2 cost map static layer update
```

9.3 Dynamic Period Adjustment

Operational Situation	Recommended Period	Rationale
Wide-area recon (open)	**5s** (default)	v1.3 shortened
General mission (default)	**5s**	Balanced
Urban/indoor penetration	**3s**	Complex terrain, faster refresh
After formation change	Immediate (1x)	Re-align with new formation
Heavy obstacles encountered	**2s temporary**	Safety

9.4 Comm Load Verification

Item	Estimated
Occupancy grid (100×100m, 0.1m resolution)	1000 × 1000 cells = 1 MB raw
After PNG compression	50-200 KB
8 robots, 30s period	Avg 30 KB/s, worst 60 KB/s
Wi-Fi 6 Mesh capacity	12.5-37.5 MB/s (uses < 1%)

9.5 Local vs Global SLAM Role Separation

Role	Local SLAM (each robot)	Global SLAM (Hub aggregated)
Update rate	1 Hz	**5 s** (v1.3 updated)
Purpose	Real-time avoidance, Nav2 local	Long-term recon, global planner
Coverage	~30m around robot	Hundreds of meters to km
Storage	Each robot	Hub UGV SBC #1
Loss impact	Reduced avoidance capability	Mission planning info loss

9.6 Local Cost Map (v1.3 new)

Each UGV computes its own layered_costmap_2d on its SBC at 1 Hz. Adopts Nav2 standard structure, and serves as input for obstacle avoidance (§6.4). Vehicle footprint + safety margin is represented as inflation, enabling the DWB local planner to safely compute ±2m lateral detour trajectories.

9.6.1 Cost Map Design Spec

Item	Value	Rationale / KPP
Resolution	0.05 m (5 cm)	235mm obstacle -> 4-5 cells
Area (footprint)	Front 7 m + rear 3 m + sides ±3.5 m	14×14 m
Grid size	280 × 280 cells	78,400 cells, ~78 KB raw
Update rate	**1 Hz** (Local)	New KPP cost_map_latency_p99 <= 5s
Coverage (forward)	**>= 7 m**	New KPP

Item	Value	Rationale / KPP
		cost_map_coverage >= 7m
Look-ahead time	2.5 sec @ 10 km/h	7m / 2.78 m/s
Inflation radius	1.0 m	Chassis + 0.25m safety
Footprint shape	Rectangle 1.3 × 0.85 m	UGV spec §4.5

9.6.2 4-Layer Structure

```
     +-----------------------------------------------+     |  Master Cost Map (10 Hz
publish, 1 Hz update)|     +-----------------------------------------------+
^                | combine     +--------+--------+--------+-------------+     | Static |
Obstacle|Travers-| Inflation   |     | Layer  | Layer  |ability | Layer       |     |
(SLAM) |  (E1)  | Layer  |             |     | 5s upd | 1Hz upd| 1Hz upd| 1Hz upd     |
+--------+--------+--------+-------------+          Layer 1 (Static):       -
SLAMAggregatedMap broadcast (Hub UGV, 5s)       - Global walls, buildings, fixed
structures          Layer 2 (Obstacle):       - Robosense E1 point cloud -> 1 Hz voxel
grid -> 2D projection       - height > 100mm: cost 200 (inflated)       - height >=
235mm: cost 254 (lethal)          Layer 3 (Traversability):       - 1 Hz ground plane
segmentation (RANSAC)       - Slope + ditch detection:         * slope < 15°, no ditch:
cost 0 (free)         * slope 15-30°: cost 100 (cautious)         * slope > 30°: cost
254 (lethal)         * ditch width >= 220mm: cost 254 (lethal)          Layer 4
(Inflation):       - radius = 1.0 m       - exponential decay from lethal cells
```

9.6.3 Latency Budget (KPP <= 5s)

Stage	Budget	Measured (PoC)
Robosense E1 -> ROS topic	100 ms	~80 ms
Point cloud -> voxel grid	200 ms	~150 ms
Ground segmentation (RANSAC)	300 ms	~250 ms
Layer combine + Master compute	200 ms	~150 ms
Cost Map publish	100 ms	~50 ms
Total (worst)	**900 ms**	~680 ms (margin 4.3 sec)

Measured latency p99 < 1 sec, with ample margin against the 5 sec KPP. The **Node Watchdog (§11.4)** detects anomalies at 3 sec threshold.

10. Video Streaming — GStreamer UDP/SRT

10.1 Change Background (v1.0 → v1.1)

v1.0 specified video streaming only as "LTE backhaul, BEST_EFFORT". v1.1 confirms:

- Encoder: H.265 (default) / H.264 (compatibility)
- Container: MPEG-TS
- Transport: SRT (operator) / UDP (inter-robot)
- Pipeline: GStreamer
- Trigger: **Android APP request (Pull mode)**

10.1.1 v1.5 Standard Toolchain (★ DCN-2026-001 D-007)

Per DCN-2026-001 D-007, all implementing partners MUST conform to the following standard OS / middleware / video toolchain. Any deviation requires DCN amendment.

Layer	v1.5 Standard
OS	Ubuntu 22.04 LTS (all UGV SBCs and operator tablet ground station)
Middleware	ROS 2 Humble Hawksbill, RMW = rmw_cyclonedds_cpp. C++ implementation only (Python prototypes deprecated per coding standard).
Video pipeline	GStreamer 1.20+ (C API only; no gst-launch-1.0 shell invocations per coding standard “Shell 호출 0 건”). Used for both SRT (operator-side) and UDP (inter-robot) pipelines.
Video codec	H.265 default, H.264 compatibility fallback. Hardware encoder on RK3588 (MPP).
Transport	SRT (operator-side, LTE/Wi-Fi 6 backbone) / UDP (inter-robot mesh). See §10.2.

10.2 SRT vs UDP Selection

Aspect	UDP (Plain)	SRT (adopted, operator-side)
Latency	Lowest (~50ms)	200-500ms (buffered)
Packet loss handling	None (visible glitch)	Auto-retransmit (ARQ)
Encryption	Separate impl	Built-in AES-128/256
Bandwidth adaptation	None	Dynamic bitrate
Firewall traversal	Difficult	Easy
Recommended use	Inter-robot lightweight	Operator screen
KPP 200ms met	Met	Met with live

Aspect	UDP (Plain)	SRT (adopted, operator-side)
		latency=120ms tune

10.3 GStreamer Pipelines

10.3.1 Sender (Follower or Hub) — H.265 + SRT

```
    gst-launch-1.0 -v \         v4l2src device=/dev/video0 \         ! video/x-raw,width=1920,height=1080,framerate=30/1 \         ! x265enc bitrate=2000 tune=zerolatency speed-preset=ultrafast \         ! h265parse \         ! mpegtsmux \         ! srtsink uri="srt://<operator_ip>:8888?mode=caller&latency=120&passphrase=..."
```

10.3.2 UDP Variant (lightweight inter-robot)

```
    gst-launch-1.0 -v \         v4l2src device=/dev/video0 \         ! video/x-raw,width=640,height=480,framerate=15/1 \         ! x264enc bitrate=500 tune=zerolatency \         ! h264parse \         ! mpegtsmux \         ! udpsink host=<hub_ip> port=5000
```

10.3.3 Hub Relay (SBC #2 forwarding)

```
    gst-launch-1.0 -v \         udpsrc port=5000 caps="application/x-rtp" \         ! rtpmp2tdepay \         ! tsdemux \         ! srtsink uri="srt://<operator_ip>:8888?mode=caller&latency=120"
```

10.3.4 Receiver (Android APP)

```
    // Android (ExoPlayer)     val srtUri = Uri.parse("srt://<robot_ip>:8888?mode=listener&latency=120")     val source = ProgressiveMediaSource.Factory(dataSourceFactory)         .createMediaSource(MediaItem.fromUri(srtUri))     exoPlayer.setMediaSource(source)     exoPlayer.prepare()     exoPlayer.playWhenReady = true
```

10.4 Video Request Flow (Pull mode)

```
    [Android APP]                [Hub UGV SBC#2]            [Target Follower]
|                             |                           |          | 1.
VideoStreamRequest       |                           |
----------------------------->|                           |          | {target_id, srt,
h265, hd}   |                           |          |                             | 2.
start_stream cmd       |          |
-------------------------->|          |                             |
|          |                           | 3. SRT stream (forward)    | stream start
|                             |<--------------------------          | 4.
VideoStreamHandle        |                           |          |
<-----------------------------                             |          | {srt_uri, key,
bitrate}     |                           |          |                             |
|          | 5. SRT receive & decode     |                           |          |
<==================================================<======| (relay)          |
|                          |          | 6. VideoStreamStop          |
|          ----------------------------->| 7. stop_stream           |          |
-------------------------->| stream stop
```

10.5 Bitrate and Quality Policy

Quality Tier	Resolution	FPS	Bitrate	Use
High (HD)	1920x1080	30 fps	3-5 Mbps	Detailed recon, target ID
Standard (FHD)	1280x720	30 fps	1.5-2 Mbps	General mission (default)
Low	640x480	15 fps	300-500 Kbps	Comm-constrained
Thumbnail	320x240	5 fps	100 Kbps	Multi-screen monitoring

10.6 Security

- SRT built-in encryption: AES-128 (performance-balanced)
- Passphrase: auto-issued at mission start (same key mgmt as DDS Security)
- Hub-Operator segment: additional IPsec or OpenVPN tunnel
- Robot-Hub segment (LAN): WPA3 + DDS Security

11. Safety + Leader Election

11.1 Modified Raft Algorithm

```
     class ModifiedRaft:         def __init__(self):            self.term = 0
self.voted_for = None             self.state = 'FOLLOWER'  # FOLLOWER / CANDIDATE /
LEADER             self.election_timeout_ms = random.randint(150, 300)          def
on_election_timeout(self):            if self.state == 'FOLLOWER':
self.term += 1                 self.state = 'CANDIDATE'                 self.voted_for
= self.my_id
self._broadcast_request_vote(my_priority=self._compute_priority())          def
on_request_vote(self, request):            if request.term > self.term and
request.priority > self._compute_priority():                 self.term = request.term
self.voted_for = request.candidate_id                 self.state = 'FOLLOWER'
return Vote(granted=True)             return Vote(granted=False)          def
on_vote_received(self, vote):            if vote.granted and self.state ==
'CANDIDATE':                 self.vote_count += 1                 if self.vote_count >=
self.quorum_size:                     self._become_leader()          def
_become_leader(self):             self.state = 'LEADER'
self._start_publish_predictions()       # FollowerTarget 10Hz
self._start_dispatch_surveillance()      # SurveillanceSector 10s
self.audit.log(category='swarm', event='election_won')
```

11.2 Leader Succession Scenarios

Scenario	Behavior	Reform Time
Battery < 30% (predictive)	Hub UGV auto-takeover (5s alert)	5-8s
Damage/comm loss	Modified Raft → Hub UGV priority 1st	5-8s
Hub UGV also damaged	Next-priority follower succeeds	≤ 10s (KPP-4)
Quorum unmet	30s wait, retry → "Re-organizing" UI msg	Up to 30s

11.3 GPS Jamming Response

Time	Behavior
0-5s	Dead-reckoning (IMU + wheel odometry), formation stable
5-30s	LiDAR SLAM augmentation (Robosense E1 + Go2 L1)
30s+	Tier 4 autonomous, return to safe area

11.4 Communication Loss Response

Time	Behavior
0-30s	Follower self-handles via Tier 1/2/3
30-60s	Auto T4 Breadcrumb entry
60s+	Auto RTH (return to home), LoRa emergency channel available

12. System Flow Diagrams

12.1 Mission Start Flow

```
    Operator (Tablet)        Leader (Go2)       Hub UGV          Followers        |
|                  |                   |        | Boot, Pair, Mesh join   |
|                   |        ------------------------->| DDS discovery    |
|        |                        | Election -> Hub Leader priority      |         |
|                  v                   |        |                          |
[LEADER or backup]      |        | Formation "Recon/Defense"|                 |
|        ------------------------->| FormationCommand                   |        |
|-Hungarian assign--------------------->|        |                          |
SlotAssignment   |                    |        |                          |
|                    |        | "Start Mission"          |                  |
|        ------------------------->| MissionStateCommand=OFFENSIVE          |        |
| Pre-check (PTP, RTK, SLAM, Sensors)    |        |                          |
FollowerTarget 10Hz start ------------>|        |                          |
SurveillanceSector 10s start --------->|        |                          | Breadcrumb
1m/0.5s start ------------->|        |                         |                  |
|        |<-- Telemetry 1Hz -------| ---- Local SLAM ->|                    |        |
RobotStatus, Formation                                          |        |
|        |   (30-60s after)        |  SLAMAggregatedMap broadcast --------->|
```

12.2 360° Surveillance Dispatch Flow (10s period)

```
    t=0          Leader: evaluate 8 robots' current poses + formation + threats
|                  v                  SectorCalculator:                    - Leader:
front 60° (fixed)                    - F1-F6: front 180° / 6 = 30° each (+overlap)
- Hub: rear 180°                    - F7: rear reinforce                       |
v     t=0.1s       SurveillanceSectorAssignment broadcast x 8 (P1, RELIABLE)
|                  v (each robot receives)     t=0.2s       Each Follower: publish
PanTiltCommand (mode=sweep)                   |                  v     t=0-10s
Sweep operation (30°/sec)                   |                  v     t=10s
Leader: recompute next sectors -> broadcast      Event (threat detected):
Detection / ThreatAlert                   |                  v (immediate)
Leader: focus 2-3 followers near threat -> broadcast update
```

12.3 Video Streaming Flow (Pull mode)

```
    Operator (Android)     Hub SBC#2          Target Follower         |
|                    |         | tap robot icon         |                    |         |
-> "View Video"        |                    |         |                      |
|         | VideoStreamRequest    |                    |         ----------------------
>|                    |         | {target_id, srt, hd}  |                    |
|                      | start_stream cmd   |         |
------------------->|         |                       |                    | GStreamer
|                      |                    | pipeline         |
|                    | activate         |                      | <--- UDP 5000 -----
| VideoStreamHandle     |                    |         |<---------------------
|         | {srt_uri, key}        |                    |         |
|                    |         | SRT 8888 connect      |                    |
---------------------->|                    |         | <-- SRT video stream --
(forward UDP->SRT) |                  (operator closes view)         | VideoStreamStop
```

```
|         ---------------------->| stop_stream         |
-------------------->| stream stop
```

12.4 SLAM Aggregation Flow (30-60s period)

```
     Each Robot (Leader + 7 Follower)        | Local SLAM 1Hz (slam_toolbox + Robosense
E1)        | Every 30-60s send SLAMLocalDelta (RELIABLE, P2)        v     Hub UGV SBC
#1        | multirobot_map_merge:        |   - Feature-based merge (ORB/RANSAC)
|   - Pose-graph optimization (g2o)        |   - Inter-robot loop closure detection
| Every 30-60s create aggregated map        | Broadcast SLAMAggregatedMap
(TRANSIENT_LOCAL)        v     All Robots + Operator Tablet        | Nav2 cost map
static layer update        | Operator UI: global map display
```

12.5 Leader Loss + Hub Succession Flow

```
     t=0         Leader (Go2) battery < 30% or comm loss     t+0.6s      DDS Liveliness
LOST callback (200ms x 3)     t+0.6s      All robots: state FOLLOWER -> CANDIDATE
t+0.7s      RequestVote broadcast with priority                  - Hub UGV:
is_hub_ugv=True (top)                  - Others: battery, health, ...     t+0.8s
Each robot grants vote to Hub UGV     t+0.9s      Hub UGV reaches quorum, CANDIDATE ->
LEADER     t+1.0s      Hub UGV starts publishing:                  - FollowerTarget
10Hz                  - SurveillanceSector 10s                  - SLAMAggregatedMap
(already on SBC#1)     t+1.5s      If old Leader still alive, demote to follower
t+2.0s      Tablet UI: "Hub UGV is now Leader"      Total reform time: ~2s (well within
KPP-4 ≤ 10s)
```

13. Traceability Matrix

13.1 KPP ↔ Design Section ↔ IDS Message ↔ Test Scenario

KPP	SDD §	IDS Message	TST Scenario
Formation hold ≤ 2 m	§6, §7	FollowerTargetMessage	S1-1, S6-2, S14-1
Reaction ≤ 300 ms	§6.2 (T1.5)	EmergencyAlert, TierStatusChange	S9-1, S14-3
Comm latency ≤ 150 ms	§5 (OpenWrt), §6.3	FollowerTarget (P0), HeartBeat	S7-3, S10-2, S14-1
Leader reform ≤ 10 s	§3, §11	LeaderElection	S11-3, S14-2
Assembly success ≥ 95%	§7 (Hungarian)	FormationStatus, SwarmHealthSummary	S5-1, S14-3
Drone detection ≥ 2 km	§4.4 RF	DetectedObject, ThreatAlert	Separate sprint
Video transmission ≤ 200 ms	§10 (SRT 120ms)	VideoStreamRequest, VideoStreamHandle	S15-5
Target classify ≥ 90%	§4.3 EO/IR, §8 sweep	DetectedObject	S15-2
Pan-tilt tracking ≤ 0.05°	§8.3 Track mode	PanTiltCommand	Separate range

13.2 v1.0 → v1.1 Change Impact

Change	v1.0 affected §	v1.1 new §	Related Messages
360° surveillance distribution	(new)	§8	SurveillanceSectorAssignment, PanTiltCommand
Hub UGV dual SBC	§3 (single SBC)	§3 (dual)	(architecture change)
SLAM 30-60s aggregation	§9 (1Hz)	§9	SLAMLocalDelta (mod), SLAMAggregatedMap (new)
GStreamer UDP/SRT	§10 (unspecified)	§10	VideoStreamRequest (new), VideoStreamHandle

Change	v1.0 affected §	v1.1 new §	Related Messages
			(mod)
OpenWrt Wi-Fi 6 Mesh	§5 (no detail)	§5	(infrastructure)
Robosense E1 LiDAR	§4 (unspecified)	§4	(sensor)

Appendix A. Related Documents

Document ID	Title	Version
SAN-PRD-SRR-001	System Requirements Review	v1.1
SAN-SDD-SUR-001	Surveillance / SLAM / Video Supplementary SDD	v1.1
SAN-IDS-CMD-001	Interface Definition Specification (35 msgs)	v1.1
SAN-OPS-SOP-001	Operator Squad SOP Card	v1.1
SAN-TST-INT-001	Integration Test Scenarios (48 scenarios)	v1.1

Appendix B. Acronyms

Acronym	Meaning
KPP	Key Performance Parameter
IDS	Interface Definition Specification
TST	Test Specification
SDD	System Design Document
DDS	Data Distribution Service
BT	Behavior Tree
RTK	Real-Time Kinematic GNSS
SLAM	Simultaneous Localization And Mapping
HFOV	Horizontal Field Of View
SRT	Secure Reliable Transport
mwan3	Multi-WAN3 (OpenWrt failover package)
DAWN	Decentralized WLAN Network (OpenWrt AP steering)
EIRP	Effective Isotropic Radiated Power
SoC	State of Charge
RCWS	Remote Controlled Weapon System